Pavan Kushal Velagaleti

Robotics Software Engineer — Real-Time C++ · ROS 2 · Embedded Control & Perception
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Education

New York University, Tandon School of Engineering — Brooklyn, NY
M.S. Mechatronics and Robotics — *Expected May 2027*
- **Relevant coursework:** RL & Optimal Control, Real-Time Embedded Systems, Robotic Gait & Manipulation, Foundations of Robotics, Math for Robotics, Advanced Mechatronics

Mahindra University — Hyderabad, India
B.Tech. Mechanical Engineering — *2019 – 2023*
- **SAE BAJA Team Captain & Head of Suspension:** led system integration and DFMEA safety analysis; mentored ∼20 members.

Experience

Articulus Surgical — Bangalore, India
Robotics Software Engineer — System Architecture — *Jul 2023 – Apr 2025*
- Architected the real-time C++ control stack for a multi-arm laparoscopic robot, achieving **20–25 ms end-to-end latency** across coordinated arms with control loops at **1–3 ms cycle times**; used mutually exclusive callback groups and no-allocation hot paths so the safety-critical path was never preempted.
- Benchmarked Cyclone DDS and Fast DDS against the latency budget (**40–50 ms with second-long spikes**, not viable), then rebuilt the transport on **iceoryx zero-copy shared memory**, cutting transport latency to **∼3 ms** while preserving the modular pub-sub architecture.
- Chose raw SocketCAN over *ros2_canopen* after testing both; root-caused silent arm oscillation to stale CAN frames feeding the IK solver and fixed it with a buffer flush on encoder desync, validated in a PyBullet digital twin.
- Developed **ESP32 (ESP-IDF, FreeRTOS)** control modules with watchdogs, fault confinement, and safety interlocks, validated through hardware-in-the-loop testing under an IEC 62304 software lifecycle.

Projects

WALL-E V3 — Autonomous Tracked Robot with SLAM, RL & LLM Commands — NYU, 2026
C++, Python, ROS 2, Nav2, RTAB-Map, Jetson Orin, Isaac Sim/Lab, Docker
- Built a tracked robot's full ROS 2 stack on **Jetson Orin Nano** (Docker Compose): YOLOv8n perception, RTAB-Map SLAM with a D435, Nav2, on-device LLaMA 3.2 command parsing, and motor control.
- Designed a safety watchdog forcing IDLE on camera dropout (>2 s), localization loss (>3 s), or serial timeout, and a sync-byte-framed binary serial protocol that recovers from dropped bytes.
- Ported the robot to **Isaac Sim/Isaac Lab** with an skrl RL pipeline (PPO/TD3); root-caused physics-fidelity bugs including a double-counted spawn offset, a leftover USD torque limit slipping the tracks, and a silently mis-pathed rigid body; fixed a *controller_server* remap silently breaking autonomous driving and added GitHub Actions CI.

ASRo — Visual SLAM Pipeline for a 6-Wheel Platform — NYU, 2026
ROS 2, Isaac Sim, RTAB-Map, robot_localization (EKF), RealSense D455
- Built a D455 + EKF + RTAB-Map SLAM pipeline in Isaac Sim; fixed a camera mounted 90° off-axis (verified via tf2) and a QoS durability mismatch silently dropping all RViz map messages.

FENCE-BOT — VR-Teleoperated 6-DOF Arm with Physics-Informed Learning — NYU, 2026
C++, Python, ROS 2, Isaac Sim/Lab, PyTorch, SymPy
- Built VR → UDP → ROS 2 → Isaac Lab teleop (C++ ROS 2 relay at 50 Hz) with DLS IK ($\lambda = 0.1$) driving joint PD actuators; fixed LNN bugs where feature scaling broke Euler–Lagrange consistency and a collision flag corrupted autodiff Hessians.

Additional Projects

- **UR10e Kinematics & Mid-Air Intercept** (MuJoCo, 2025): DH FK, multi-start numerical IK, Jacobian velocity kinematics, and a task-space PD torque planner that intercepted **all 6** ballistic targets.
- **Vision-Controlled Robotic Finger** (2025): MediaPipe hand tracking → UART → encoder-feedback PD loop on a tendon-driven finger, with the plant modeled and stability validated in MATLAB/Simulink before hardware.
- **PX4 Offboard Autonomy & Home Service Robot** (2024–25): ROS 2 offboard missions over PX4 SITL with ArUco-triggered phases; Gazebo delivery robot with RTAB-Map mapping, AMCL localization, and costmap navigation.

Technical Skills

Languages & Tools: C++ (real-time-safe), Python, C, MATLAB/Simulink, Bash, Git, CMake, GitHub Actions, LTTng
Robotics & Sim: ROS 2, Nav2, ros2_control, Isaac Sim, Isaac Lab, MuJoCo, Gazebo, PX4, CAN/SocketCAN, iceoryx, DDS
Perception & ML: RTAB-Map, EKF, YOLOv8, OpenCV, MediaPipe, ArUco, PyTorch, skrl (PPO/TD3)
Embedded: ESP-IDF (ESP32), FreeRTOS, STM32 HAL, Arduino, Jetson Orin, Docker